```
function coinToss(): bool
{
    return (bool) rand(0, 1);
}

$msg = "Today I shall ";
$msg .= (coinToss()) ? "procrastinate." . PHP_EOL : "work on my book!" .
PHP_EOL;

print $msg;
```

The function `coinToss()` returns `true` or `false` depending on the whim of the machine. It might explain why this book took me so long to write.

PHP 8.0 introduced a type declaration that also checked the data's value by allowing `false` as part of a union type. The argument for this was that historically many PHP functions return false on failure, so supporting user-defined functions with this type declaration made sense. For example, the built-in function for reading in an entire file has the following signature: `file_get_contents(): string|false`. There are many, many more built-in functions that follow this false-on-failure pattern.

PHP 8.2 added support for the complementary `true` type and did away with the need for unions; `false` and `true` can be used as standalone types. The benefits of allowing standalone `true` or `false` over `bool` aren't immediately apparent when examined from a purely functional perspective: why would we write a function that can only return `false`? In the context of objects (and especially class hierarchy), however, there are use cases that emerge. A parent class could provide a prototype method with a return type of `bool`; child classes can then overload this with `true` or `false` return types and provide more granular detail to your IDE or static analysis tools. If that last sentence was just so much gobbledegook, other chapters in this book have all the info you need.

A word of warning: don't try to use a `true|false` union. PHP will quite rightly tell you to use `bool` via a helpful fatal error message!

The `mixed` (Super) Type

PHP supports a type declaration of `mixed`, which is the top-level supertype (in that all other types are a sub-type of `mixed`, for the type theorists among you). The `mixed` type is (according to PHP official documentation) equivalent to the union `object|resource`